Key characteristics of blockchain technology

Every transaction that records and stores is not labelled as blockchain. The following are the main characteristics of blockchain.

- **Digital:** Digitises all the information, thus eliminating the need for manual documentation.
- **Distributed:** Blockchain distributes control among all peers in the transaction chain, creating a shared infrastructure within an enterprise system. Participants independently validate information without a centralised authority. There is no single point of failure because of this. Even if one node fails, the remaining nodes continue to operate, ensuring no disruption.
- **Immutable:** All transactions in the blockchain are immutable. Encryption is done for every transaction covering time, date, participants and hash of the previous block.
- **Chronology:** Each block acts like a repository that stores information pertaining to a transaction and links to the previous block in the same transaction. These connected blocks form a chronological chain providing a trail of the underlying transaction.
- **Consensus based:** A transaction on blockchain is executed only if all the parties on the network unanimously approve it. Also, consensus based rules can be altered to suit various circumstances.
- **Digital signature:** Blockchain enables exchange of transactional values using unique digital signatures that rely on public keys. Public keys are decryption code known to everyone on the network. Private keys are codes known only to the owner to create proof of ownership. This is very critical in avoiding fraud in records management.
- **Consistent:** Blockchain data is complete, consistent, timely, accurate, and widely available.
- **Persistence**: Invalid transactions are not admitted. It is nearly impossible to delete or roll back transactions once they are included in the blockchain. Cryptographically, the blocks created are sealed in the chain. It is impossible to delete, edit or copy already created blocks and put them on the network. This ensures a high level of robustness and trust.

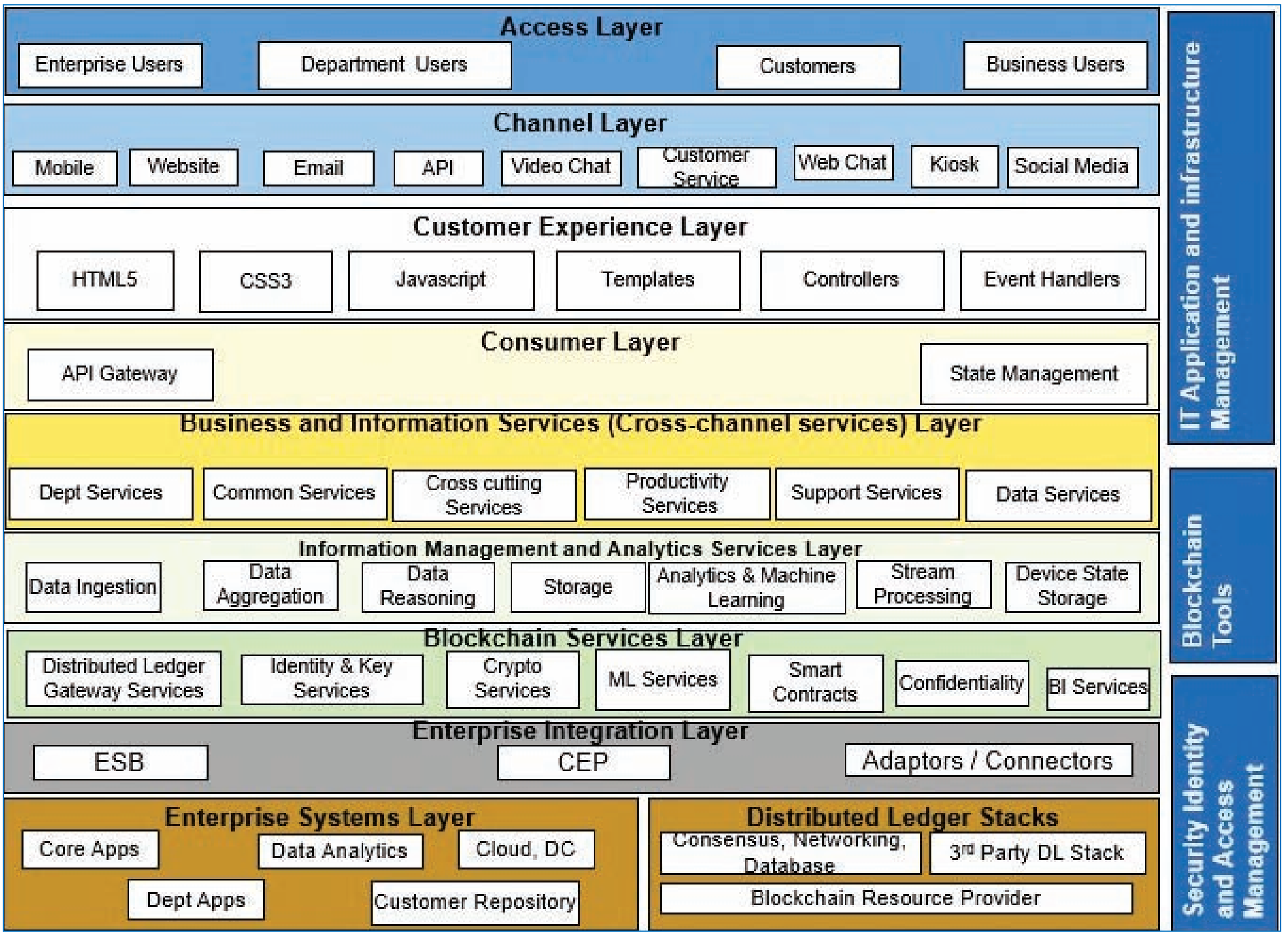

Figure 1: Blockchain logical application reference architecture